

Active Inference BookStream #001.01 ~ "Governing Continuous Transform

BookStream | 1:00:27

foreign

it's December 6 2022 we're in active

bookstream number

001.01

a numbering convention that will be

explained soon

it's a discussion on the book governing

continuous transformation by vijayan

kazri and I'll pass the blue

hi everyone I guess that's my job is to

explain the uh bookstream numbering so

this is our first book stream that's why

it's bookstream001 and this is our first

dot zero contextualization video for

this bookstream series so each week we

will present a chapter ish a chapter or

so of the book and then periodically we

will get together for the zero zero one

dot one zero zero one dot two and so

forth those discussion sections so if

you want to participate either in a

contextualization video or in a

discussion video please get in touch

with us and we would love to have you

um on the stream and I'm blue I've been

here before so hopefully you've seen me

before and know me by now and I'm an

independent researcher in New Mexico and

I will pass it off to Tyler first time

appearance I think in the active

inference Institute so Tyler do you want

to introduce yourself yeah cool thanks

for having me this is my first time here

to actually inference so I have a an

unusual background actually don't come

from in an academic background at all a lot of it's been about applied governance so I actually started as a strategy consultant working in major corporations more recently it's gotten more to Dow governance I currently lead product and protocol design for the 4K protocol it's a project that is connecting the physical world to the on-chain world and I'm also a contributor at medigov where I think about how we can model these abstract formalisms about governance in a real world setting and so a lot of times when I think about Dow governance but most recently it's been thinking about active inference and implicit implications for real world settings

foreign

so we are a participatory online lab that is communicating learning and practicing applied active inference these are all of our media links um and we will be recording And archiving this live stream for future reference

um everyone is welcome and hopefully we will be following good etiquette for live streams and not arguing or talking over each other for the duration of the book streams

um

okay so this is our first book stream um our goal is to learn and discuss this textbook governing continuous transformation reframing the strategy governance conversation it's a 2022 book by Bijan kesri and this video like

always is just an introduction to some
of the ideas not a review or a final
word at all so we will just go through
here

um the sections of this book stream
we're going to do the book map and the
road map some keywords the preface part
one and chapter one will be covered in
today's video and in the coming weeks we
will be continuing to discuss this so
get in touch if you want to partake in
these conversations

um so I guess we introduced ourselves
briefly

um oh backwards so if either you guys
want to maybe mention something you're
excited about or that you liked or
remembered from the book from these
first sessions

um otherwise we can go on to discover it

I think we have a lot of material Tyler

what are you most excited about

yeah I mean I think for me like

especially being in like real world

organizations non-profits a lot of times

we think of like uncertainty as

something that could be wished away or

something that we're incrementally going

to solve like oh if we only were

collecting like more information we

would be in a better spot or I don't

think we had more capabilities this we

wouldn't have this problem

um but a lot of times we kind of get

very anxious about that uncertainty

rather than we're an active inference

and free energy governance that

uncertainty is placed at the center and

it's something you could actually take ownership over but also be a source of you know creative uh growth and so I think that's really exciting and inspirational and yeah really excited to be here today

cool I'm excited too to unify some of the concepts of active inference with governance which I know a little bit about but not very much and now that we're like formalizing our structure as an Institutional entity in the active inference Institute

um I'm excited to learn more about governance and how things should work um okay so this book has four parts so we're here well we're not even in part one yet but we're going to go over to the beginning of part one today so part one is reframing the strategy governance conversation part two is about free energy governance and part three is a conclusion for the book and then there are some interviews in part four along with acknowledgments and an epilogue and this is from the bookmap software so this is like our road map for a book um but this bookmap software is used in trading so if you guys have never seen it that's what these graphs are from so when I think about bookmap I always think about that uh okay so our roadmap for today so we're going to get into the preface there are no section headings so we just kind of chunked out like what makes sense

um in terms of of the text and how it reads so we're going to go over a

summary of the fep how to ground firms
in the framework of active inference
strategy governance model models versus
meta governance the triplet structure of
free energy governance and then realists
and anti-realists in management science
and then just a brief intro to part one
and then chapter one which is pretty
short thinking born of curiosity revolt
and change

um so here's like a huge list of
pre-words of pre-words of keywords that
we're going to get into today and I
think Tyler is going to start us off
um in we're in section one in the
preface in the summary of the free
energy principle Tyler's gonna start us
off talking about the fep

Tyler and mute

sorry uh so the pre-energy principle or
fep as we're going to be calling it uh
this is John's definition that he gives
in the preface so the fep uh is a
mathematical formulation that explains
from first principles to characteristics
of biological systems that are able to
resist Decay and persist over time it
rests on the idea that all biological
systems instantiate a hierarchical
generative model of the world that
personally minimizes its internal
entropy by minimizing free energy
blew anything you'd want to add there
no I think um it's best like we could
take a deep dive into the fep there are
many live streams where that's done so
if you want to learn more about the free
energy principle I would just refer out

to other live streams it's best here to take what Bijan presents the aspect that he's using and go forward from there cool and I think uh similarly we'll do a similar cursory treatment of Markov blankets uh and so markup blankets so we could think about it is that it's a statistical formulation for defining the internal versus the external so the definition that he gives is Marco blankets establish a conditional Independence between internal and external states that renders the inside open to the renders the inside open to the outside but only in a conditional sense so I.E the internal states only see the external States through the veil of the Marco Blanca again huge topic we're not going to go super into it unless you're gonna add anything blue no it's great um and and there are many many live streams like really if you want to get into it probably the emperor's new Markov blankets is maybe where we undertake like a very thorough um discussion of the Markov blanket if you want more information there all right so grounding firms in the framework of active inference so here's a claim from the book learning to actively infer will determine the firm's competitive advantage and survival in a discontinuous and distributed world the more fluid the boundaries between firm and Eco Niche the more important the definition of what constitutes a system and how non-discipitative markovian boundaries

preserve the firm's Integrity I.E belief
systems our interest is Shifting away
from pursuing deterministic truth
through clever analysis towards
designing entailing a generative
self-organizing model that empowers the
firm to master the shifting conditional
independencies between the known and
unknown and what lies within and Beyond
its control whoa so there's a lot there
so I think these next keyword slides
hopefully will unpack it a little bit
uh active inference
um which we've seen many times in the
active inference Institute is a process
Theory grounded in the free energy
principle that explains Behavior
perception planning and action in terms
of probabilistic inference
um Fijian doesn't yet mention active
inference but he talks about learning to
actively infer which I'm going to
actively infer means active inference
and here we see this classic action
perception Loop where we have internal
States and external States separated by
a Markov blanket or a non-disruptive
markovian boundary and we have sensory
input and action and these are the ways
in which an agent can interact with this
world do either you guys have anything
to add here

Daniel

um in that framing I found it really
interesting when there's increased
fluidity or fuzziness or blurriness of
the engagement of the firm with the
niche or the context or the community in

those cases it's like more important to have a uncertainty oriented definition of what the thing is and that is what confronts every organization today yeah interesting

um okay so he talked a little bit about shifting away from deterministic truth and so like I I like Googled because I'm a truth a truth person I'm weird about truth if you've ever like listened to me talk about it before so deterministic truth is like the truth

um and I I liked this paper like I just Googled it like what do people talk about what is deterministic truth

um and I I found this paper approaching determinist deterministic and probabilistic Truth a unified account and and in this paper it's a 2021 paper so pretty new

um in this paper they talked about the truth itself like the truth being either deterministic like I think about that like hitting a bullseye like the truth is the number seven

and then probabilistic truth is more like a probability distribution around that truth like hitting seven plus or minus the standard deviation or like a probability density right

um and this paper talked about like deterministic and probabilistic Truth like that the truth is one of those things and then also there are ways to approach it like you're approaching your search of the truth in either a deterministic or probabilistic way so I thought that this was an interesting

read and kind of if you are interested
in delving deeper into like
deterministic deterministic versus
probabilistic and these theories and um
approaches and and Truth likeness I
thought this was a cool um read
yeah I think one thing oh sorry no the
whole thing to add there is just I think
like our personal ideas about what truth
are like whether it's deterministic versus
a probabilistic conception of Truth
really carry over onto the organization
level and so whatever kind of delusions
you have about your personal truth in
your personal life a lot of times they
view the organization as like the CEO
abuse the organization as an extension
of their own self and so their own world
view gets carried over and you know
strapped on to the organization of the
world view

um and so it's really interesting that
you could kind of frame these like
philosophical ideas about truth that are
often applied at an individual level and
then kind of like scale that up into an
organization

yeah very cool

um and we see that in this um this
generative self-organizing model again
this is kind of like it was cool to just
search up what is a self-organizing
model

um and and we see generative Model A lot
um we don't really see self-organizing
model we see generative model in terms
of self-organization but a generative
self-organizing model like what is this

really

um um and so just this is just from this

um Stanford

machine learning cheat sheet supervised

learning cheat sheet so here it says

discriminative models divide the data

space into classes by learning the

boundaries and this is like the

different neural networks whereas

generative models understand how the

data is embedded into the space like an

auto encoder or boltzmann machine or

self-organizing Maps so this is like a

self-organizing model and you can really

see here like this is the like a

decision boundary on the left in the

discriminative model and like either

it's a cat or a dog right and on the

right it's there's a generative model

like it's a cat with some degree of

certainty or it's a dog with some degree

of of certainty and so you see these

like probability distributions like

probably it's a cat but then there's

like some fuzzy like if your data point

lands like a little bit in the red or a

little bit in the blue like then then it

might be a cat or it might be a dog

maybe we don't have enough information

to tell

um so I thought that this was a cool

like replication of the probabilistic

truth versus deterministic truth like

either you're looking for cat or dog or

you're looking for with what probability

is this a cat or a dog

um do you guys have anything to add here

cool okay so this is moving on out of

this claim into the next claim

um The Firm is envisioned as an inactive

sophisticated and resource dependent

inference machine in form of a

generative learning model essentially it

is a statistical model linking in

circular causality action perception

firm resources and the unknown external

world and I just put this active

inference Loop up here again so you have

like the firm and the world the firm

would be the agent and then you have the

resources of the firm which would be

like the internal States and then the

external world

which is everything that's not the firm

um and I highlighted these terms

inactive and sophisticated because

there's a lot to unpack here

um and I think it's kind of in the

textbook maybe like just you can say the

words but unless you like open them up

and look what they mean inside look

inside of them like you don't have any

idea the kind of depth that goes behind

these things

so

um do you guys have anything to add

there on that okay

um inactivism is a position in cognitive

science that argues that cognition

arises through a dynamic interaction

between an acting organism and its

environment and so these are the four

e's of uh cognitive science and so or

cognition and they're embodied and

active extended and in cultured so

cognition is grounded in our senses and

concrete experience that's embodied
enacted means it's goal directed action
in the real world

um

extended is cognitive systems include
the tools the devices the people around
us our smartphone these kinds of things
and embedded or that's kind of what we
always refer to as in cultured but I
think that they kind of mean the same
thing so this is cognition is woven into
culture and learning is done in a social
context so these are the the embodied So
when you say the firm is inactive

um yeah that that's it it's good for
goal directed it's doing goal directed
action in the real world and active
and sophisticated or do you guys have
anything to add there on the 40s okay
sophisticated is from this paper I
believe a sophisticated inference by
Kristen and colleagues and we've done
sophisticated affective inference I
think on a live stream but I don't think
we've ever done sophisticated inference
but the affective inference is
um like a later paper and covers this
one a lot so sophisticated inference is
really like in deep time right so here
I'll just read this the figure caption
so I have this like decision tree model
a picture from the paper and it says
this schematic summarizes the
accumulation of expected free energy
over paths and trajectories into the
future this can be construed as a deep
tree search where the tree branches of
reliable actions at each point point in

time and the likely outcomes consequent upon each action so this is one two three four time steps here and going off so this is like a trajectory the path of least action uh so we talk about you know the way that

a system can decide and go forward in in space

so that's what's meant by sophisticated I think

oh Daniel

yeah

the purpose of introducing these terms from cognitive sciences and Neuroscience like the 4E cognition and of this type of model it's not a novel idea to make a decision tree that's how for example early chess playing artificial intelligence's work

it's the way that the expected free energy can be calculated across these different Paths of consideration with respect to a generative model that are allowing new ways of thinking about these kinds of deep time and nested strategic questions

um okay

so temporal thickness and counterfactual depths here in this claim Bijan says the proposal on offer here is that the Mind comes into being when self-evidencing has a temporal thickness models can be thicker and thinner deeper or shallower depending on how far forward they predict or counterfactual depth how far back they post it that is whether they can capture how things might have ended up if they had acted differently there

is no real reason for Minds to exist
they appear to do so simply because
existence itself is the endpoint of a
process of reasoning Consciousness I'd
contend is nothing grander than
inference about my future and in this
this is from bhajan's book but he's
quoting friston 2017 paper
um so that temporal thickness goes back
to this sophistication right which is
what why like so here you can see how
how far back you can post it like if I
had done something differently what
could have happened
um this counterfactual reasoning and
then and prediction going forward into
the future
okay going on to strategy governance
models versus meta governance
all right so I'm kind of out of my uh
Comforts I'm here so I might call in
Tyler to help me out a little bit
um but Bijan says the common
understanding underlying entrenched
strategy governance models rather
erroneously has been that action is a
consequence of perception and therefore
strategy must be environment driven so I
look up strategy governance model and it
gave me like some figures and this is
one of them and it appears to be um
pretty linear but but there's not a lot
of feedback happening with the external
environment like there's Vision the
strategy a road map design and then
insights come forward to the design and
back to the division and then there's
like a performance review so I don't

know Tyler do you want to give any additional comments here on this yeah I mean I think something that could be potentially confusing for people who are maybe not coming from an academic background is that like a lot of times what Business Leaders are thinking about how they're going to plan for the action that their organizations to take what the strategy looks like you know they're getting data right and they're having feedback from the bottom levels of the organization feedback to the top levels right and so that they're taking information they're creating a perception of that and then they're creating a strategy based on that that kind of in a linear fashion the organization executes and at some point they reflect back and be like hey did this work or not what do we do with our new information right but really what they're not realizing in the process of that is like in that linear flow they're not recognizing how they're constraining the perceptions and actions of their organization in a top-down fashion and allowing and allowing bottom-up information to change how information and actions can be changed in the future so like that kind of cyclical nature really to understand it's really more conceived in this linear way of like hey we're going to collect as much information as we can our information is going to be imperfect but we're going to get better over time rather than like hey fundamentally our

actions and perceptions are constrained by the actions we're taking right now so it's like just a different model for thinking but it's a lot more linear and not doesn't have that cyclical causality uh baked into it

cool and I think about here like um it seems like it's very like um sequential like one after another after another after another whereas like the active inference kind of method like you have to sometimes act in order to perceive right like the if you want to know whether like the surface is rough or smooth you have to move over the surface with your finger to determine if it's rough or smooth so it's action and perception that are happening simultaneously and I think that that that is the shift that Bijan is calling for

um as he says

yeah and I think just to add in that like a little bit is that I think a lot of times Business Leaders conceive of like reality as like independently existing out there and it's just like a limitation of their tools and resources to understand reality rather than like the nature of perception itself of being able to understand reality and so I think that's really like the fundamental difference and a lot of times people don't realize that that's even like relevant at all uh and so that's why this is fun to talk about

that's for sure why like what is deterministic truth like where is that

reality

um and it all kind of links together

um so Jean goes on to say in fact action

upon the firm's Eco Niche akin to

pointing the light torch in the dark

determines perception inference is

inactive the generative model itself

never encodes anything it just expresses

the circular causality among all four

states the generative model hence is a

model of the firm's econish if not it is

no good regulator and consequently The

Firm will rather sooner than later

dissipate in the face of entropic forces

um and so I included here A different

kind of strategy governance model that's

like the Lean Startup model that's like

a little bit more action and perception

happening in like a more simultaneous

way I don't know if you want to add

anything here Tyler

yeah I think um a lot of the way we talk

about like Lean Startup model of really

the least our model is basically like

you don't necessarily know what people

want and also it's not that you're

discovering what people want is that

you're putting ideas out there and

finding how people respond respond to

them uh and so it's basically truth is

something that you create you can create

a new state of the world rather than

like uncovering it so it's kind of like

a method for uncovering something that's

going to work and a lot of times it's

kind of done how can like very like

fluffy and also just like experiential

terms of like hey this works we've done

this we built this company this way but
what's interesting about active
inferences that can take that very like
fluffy wispy framework and apply that
into a more rigorous context uh so yeah
like a lot of these ideas that we're
talking about now like they're not
necessarily like new and like especially
like startup world but there hasn't been
great language for talking about it
cool

all right so here's another claim
therefore the firm's true governance
challenge is a matter of meta governing
its generative model as an enacted
sophisticated and resource-dependent
belief system exercising the entire
organization quote the body itself in
Contours we should now start seeing a
model emerging that underwrites a form
of Auto autopoiesis I.E

self-organization effectively it is all
about casting inferences optimization
mathematically by way of gradient
descent starting with two essential
constituents first a hyper prior which
is the firm's resistance to entropy in
order to persist and two epistemic
foraging and hypothesis testing I.E
acting upon the Eco Niche to generate
belief evidencing data

um do you guys have any comments here do
you want to delve into this a little
more Daniel

yeah one interesting uh usage of a term
there is that the firm's true governance
challenge is meta governance and that's
a term that is used differently or

compatibly but with a different focus in
medigov and so just to kind of Flesh
that out here meta governance is being
used in the sense of metacognition like
self-reflection memory anticipation
attention planning all of these
individual level cognitive and
metacognitive processes that we model in
active inference

often medicov in another sense is meant
like social cognition like relationships
amongst organizations or amongst teams
so like within a team you have their own
governance structure then also there's
teams of teams and that's a meta
governance structure with respect to the
team

and then that's very interesting because
the two times that we've seen nested
models nested generative models inactive
inference are the metacognition models
and then the ecosystem nested models so
it really brings Integrity between
thinking about what happens like within
and across groups

cool thank you

um okay so hyper prior

um in Bayesian statistics a high

proprietor is a prior Distribution on a
hyper parameter that is on a parameter
of a prior distribution as with the term
hyper parameter the use of hyper
is to distinguish it from a prior
distribution of a parameter of the model
for the underlying system

so like the hyper parameters influence
the prior here and like the hyper
parameters here influence the parameters

here and so it's backwards at least
one step in time

in this sophisticated format so there's
oh yeah this is a better diagnosis or
illustration the hyper prior goes to the
prior goes to the data so it's just in a
previous time step in some way

um anything to add here

all right

epistemic foraging I pulled this from
active inference and learning this this
quote

um it says inactive inference Behavior
has explorative epistemic and
exploitative pragmatic aspects that are
sensitive to ambiguity and risk
respectively where epistemic Ambu
ambiguity resolving Behavior enables
pragmatic reward seeking Behavior and
the subsequent emergence of habits
um and and I just wanted to kind of
point out here that that active
inference is kind of like has been
Illustrated as like a balance a way to
balance exploration and exploitation and
epistemic and pragmatic value
um and like in business is this like
knowledge and and money is that like
epistemic and pragmatic value like
knowledge of your customers your clients
so it's just trying to think of like how
this might um translate to that like
like money pragmatic values like more
research sources and you know like
epistemic value is like more knowledge
of your client base or or of of the
future maybe
um uncertainty production

and I could just give like a concrete example of that because like it was especially true in software engineering where that is actually frame and explicitly these terms and uh like growth organizations so if you're on the growth team you're like okay we can either like turn up the dial on this thing that's working we're like okay like we know this flow is working this wizard for this checkout flow is working let's make it like 10 faster would be an example of exploitation where like we know that's going to work in some level where acceleration could be something we have like really no idea this is like a shot in the dark let's figure out the minimum amount of effort we can put out to to uncover new territory and so like this exact trade-off is very much explicitly thought of in a lot of tech organizations very cool thank you yeah so epistemic foraging is like you know searching around so like you know for for epistemic seeking out um epistemic value um okay so moving on to the triplet structure of free energy governance uh Tyler this is for you I think um so you know Jean's frames like really there's like a key question of your energy governance when you're thinking about your organization how do you embody the academic emergency organization

um and it's really about how you create
and update your generative model so that
the firm can survive and he presents
this framework of thinking about it and
he calls it the triplet structure of
re-energy governance there is structure
cognition and then capabilities so we'll
go through uh each of these in turn
all right structure

so oh sorry this is confusing B
that header is wrong just get to the
next slide

all right structure

um so this John writes that connectivity
emerges in flow and language bottom up
stimuli emanating from resource markets
such as clients Technologies suppliers
another are only as relevant and
meaningful to Resource configuration
asset allocation as clearly as top-down
predictions and prediction models
articulated communicated and above all
programmed to process bottom up incoming
signals

um so like we kind of touched on this
earlier but like a lot of times decision
making at organization is very linear
where like at the very top you're coming
up with a strategy and then like that's
executed in current at every rung of the
hierarchy and a lot of times like okrs
if you've ever heard of this it's like a
really good example of this it's like
objective key results where like there's
an objective that the top the
organization comes with and then like
the minions the next run have their sub
subsequent parts of that tree and it

kind of filters down to the rest of the organization

um where sometimes what's not as clearly articulated is about the interconnections between different parts of the hierarchy and how those different parts of the stack uh have to relate to each other so like a really good example of this is that you know this is very true in tech companies tech companies collect a ton of data and none of that data is like ever processed in a meaningful way and so like a lot of times you'll say like oh like we got we're data driven company we collect all this information but in reality like a lot of information is trapped at the bottom rungs of the organization it may not have like the political ability or also even the technical ability to process that in a meaningful way and bubble that up through the organization and a lot of times there's a huge gap in being able to actually process that information and bubble it up appropriately uh in a way that can actually like crater the organization and so this is like designing this in a really thoughtful way is much harder than it would seem and even if you call yourself a data driven organization there's a good chance you're not processing data in a way that can actually be generate new top-down models later on so yeah anything to add there where Virginia

all right let's go to the next one

cognition

all right so cognition Embraces the environment as enacted through relations and interactions constructed in our brains environment is not objectively accurate independently exists but is revealed by a firm's actions and perceptions that those firms that those very actions generate in fact the consequences of the firm's activity construct it's equal niche um and so we've already kind of touched on some of the earlier slides But the A lot of times these firms conceive of external reality existing independently of their own actions and perceptions and then it's just like up to them to get better tools and information um to be able to like understand that reality like understanding this fog of war that they're like slowly uncovering um but they actually do real insights then also the actions that they're able to take so like to make this like a little bit more concrete you can say like I mean if you're not this very much applies in a real Sense on like a website where like if you're not collecting information on views on your page in a very basic level it's really hard to like generate actual information out of that right and so you have to basically make a decision of like this is the information we want to gather this is what we care about and that's going to structure the insights you can glean in the future and like what you can do with that information in

a similar way like if you're not putting resources out for interacting with the world like creating a new Department that's maybe a little bit more exploratory that's going to constrain the actions you can take in the future so all this is saying is that uh the reality doesn't exist independently of you it's something that you actively perceive through your actions and perceptions down blue and add anything there is it like actively perceived or actively create um actively sorry I actively create a better way of saying it well or or both I mean I think like you have to be um either actively perceiving but but um I don't know I think it's it's a rare treat when someone takes responsibility or even like a company takes responsibility for um the reality that they're creating um and like we see this especially like in tech companies like I don't know Facebook and I mean like just like a lot of the technology has been life-changing and world changing so like they're really actively creating reality Daniel yeah this is uh interesting about the revealing of the firm's generative model through its actions and we've talked about that four-fold partition with the internal external States and then the sense and action blanket States so the firm has access to sense and internal states to some extent but the world or others have access to

one's actions and or
inference on their consequences in the
niche and so it's a way to talk about
what the entire action perception Loop
is with all of its different inward
facing and outward facing Dynamics
yeah to add a little more here too like
there's a lot of focus on around this
idea as well and like startup culture
and so I I'm a totally butcher there's
like a Steve Jobs Club totally butcher
about something around the lines of like
if you had done a marketing survey of
people before they made the iPod and
asked them how they want to listen to
music they wouldn't have told you that
they wanted an iPod they would have told
you something else that was conditioned
based on their previous experiences
products and so the idea is that you
just put something out in the world it's
a creative act and that you create
something in the world and then the
world responds to it
um and so there's like this big idea
especially in like Tech culture around
like reality is something that you will
into existence rather than
um discover independently
yeah it's interesting too to like think
about I mean especially you think about
Google Glass that came out like I don't
know 10 years ago or something now I'm
probably not that long ago but it's been
a little while and I think like now
there's like the re-release of like
Google Glass or some kind of similar
product like the world just wasn't ready

when the Google blast version came out
but but like we'll be ready soon we've
they've wielded into existence and it'll
be that way soon
all right capabilities the last one is
capabilities
um so the inferring firm is constantly
optimizing belief propagation as well as
updating beliefs uh sophistication that
process that process of optimizing
recognition densities is only as good as
the firm's General the world model so we
are no longer just interested in the
actions consequences per se uh about how
the actions consequences impact
ulterior beliefs
um so
another way of saying this is that the
firm has to actively balance you know
this exploiting for reverse
um exploring
and I think there's like a lot of again
folk wisdom and especially in Silicon
Valley culture around this trade-off
whereas like a lot of times we say hey
failure is good as long as you're
learning and integrating those those
insights
um and so
yeah it blew anything to add there
no good all right
um okay so realists and anti-realists in
management science
all right so here's a claim from Bijan
management science is equally divided
along not too dissimilar realists and
anti-realist fault lines it is best
captured by Carl week 1977 when he

states that environment is located in the mind of the actor and is imposed by him on experience in order to make that experience more meaningful. It's seldom Dawns on organizational theorists to look for environments inside the heads rather than outside of them. Indeed, the rise of cognitive lens in strategy management and governance research has enriched our understanding around improving the design of strategy governance processes but has largely remained constrained by the notion that environment is something objective out there waiting to be analyzed and adapted to.

um and in in this section uh Bijan talks about Blitz scaling and um mentions some like examples of of this anti-realist uh claim so he says Google Uber Airbnb and Facebook are the product of a fundamentally anti-realist enactment of an interaction with the respective Eco niches powered by unique and continuously evolving sets of capabilities fostering action generated top-down bottom-up prediction error minimization to iteratively optimize their very generative models and implied recognition posterior probabilities of beliefs about the actions consequences these companies redesign while they fly which is in the words of Silicon Valley veteran Reed Hoffman equivalent to throwing yourself off a cliff and assembling your airplane on the way down Sullivan 2016. in fact we are more in control than we resist to believe Blitz

scaling has been integral to the founding DNA of these super outperformers they listen to incense what wants to emerge they are the enacting designers of their very Eco Niche for these companies strategy is rather a matter of ex post sense making then ex-anti long-term planning it is a matter of survival

and I just thought that that was maybe a good example of this anti-realism um thing that that he was talking about here

uh do you guys understand an idea that kind of arose as you were talking about this is that one interesting point about Google is that they're also kind of famous for trying really hard to do this but failing to create new products so they've like created and killed like hundreds of different new like products that people genuinely like truly loved even that were quite Innovative and so it's kind of interesting of like how you even think about like

what is trying to emerge and whether that's best done in like a single monolithic organization versus in a more like decentralized distributed way um so I don't know I this has been like a tension I've been kind of feeling throughout the book as well it's kind of like what is it that this entity is trying to exist or it needs to exist or is it something that is more distributed and decentralized is more appropriate in certain cases

so having loved and lost several Google things in the past that have like people are either uh yeah Google or Google App Maker like the low code app developing app development thing I mean just like tons of things right having loved and lost I wonder if this is like epistemic exploration on the part of Google really so so if like you're you know I don't know I feel like people come in and like I'm gonna build this app and Google gives them a lot of freedom and they're like go ahead build the app or build the platform or build the thing and do the thing and they build it and it's great and people love it and then they move on and do something else and then nobody steps up to support the love of this part or what was this person's like child at Google right so I feel like maybe that might be a lot of what's happening but but perhaps in this like creation and then unsupporting creation and unsupporting Google is continuously generating like epistemic knowledge right it's epistemic foraging of and learning about the niche and what people like and what they will use you know it it like on the grand scheme well I think it's I think that's true but I'm not going to make this argument but a lot of people have made the argument that what they should have been doing the whole time is exploiting because like their home business is just ads and they're like just like run that until you die like that's the only thing that matters

like your resources aren't well used
here by like trying to do like all of
these random things
um and so like there is like a
legitimate argument too of like the
limits to which you can explore maybe I
don't know that's probably out of scope
for our discussion today but
um all right Daniel yeah just one short
note the entire discussion of epistemic
and pragmatic value is motivated
motivated by the fact that they're
unified in the free energy functions
that are used to determine perception
cognition and action in an integrated
way so at the very least conversation
can be had about adaptive changes in
these variables at different levels of
the organization
in a different if not better way that's
very integrated with how we think about
the world internally and built upon
Neuroscience in that area as well as
about thinking about the complex
Dynamics and the relational dynamics of
the niche which for information doesn't
exist until you're seeking it out and in
feedback with it and inattention of your
previous data in anticipation of future
data all of these cognitive Dynamics so
the anti-realist take doesn't need to be
someone who doesn't think like a real
world exists or anything like that even
though that's like too deep for this
current discussion it's just a purely
pragmatic approach to thinking about and
modeling these kinds of scenarios
very cool all right what else ah okay so

here we are we got through the preface
that was the preface
um and so getting into part one of the
book uh there's just this small quote
which I'll read the it says reframing
the strategy governance conversation the
evidence suggests that we are in the
midst of an evolutionary punctuation we
are witnessing a mass extinction in the
corporate world in the early Decades of
the 21st century since 2000 52 of the
Fortune 500 companies have either been
acquired merged or have declared
bankruptcy it is estimated that 40 of
the companies in existence today will
shutter their operations in the next 10
years merely following the trends of
change is not enough just like organisms
facing the great oxidation event
organizations need to reinvent the way
they interact with the changing world
and this is a quote from digital
transformation survive and thrive in an
era of mass extinction at 2019 by siebel
all right chapter one thinking born of
curiosity revolt and change all right
here's our abstract Tyler I'm gonna let
you read this one
all right environmental complexity and
velocity have been dominant moderators
and strategy and governance research
the underlying structure of
environmental change is becoming
increasingly distributed and
discontinuous the disciplinating nature
of Information Technology is challenging
the trench business models as well as
the Logics of organizing socialized

top-down strategy processes are rather
anti-clocked unless dynamically and
generatively intertwined in real time
with bottom-up stimuli and data
emanating with field operations in
pursuit of prediction error minimization
neither established Logics of organizing
or prevalent corporate governance models
are a match for wholesale digital
reinvention and continuous strategic
renewal

nice and going on he says

in fact akin to the AI Evolution from a
supervised to a deep learning logic
corporate governance is challenged to
evolve from a supervised and rule-based
best practices Learning System to one of
unsupervised unlearning I.E

self-organization

to survive the firm must be conceived as
one cross hierarchically integrated
inference machine organizing is
generative prediction error minimization
strategy is redefined as top-down
bottom-up predictions processing the
search for knowledge is nourished not
nourished by certainty it is nourished
by a radical absence of certainty
thinking born of curiosity revolt and
change and that's a quote from roveli
2021.

all right all right Tyler

so uh one of the core premises of the
app the abstract is saying hey the
environmental complexity and velocity
increasing over time and that makes it
more challenging for organizations
um to survive so what he says is

environmental complexity and velocity have been dominant moderators and strategy and governance research indeed the rate of change has not only been accelerating but the underlying structure of change has become increasingly distributed and discontinuous sense of urgency and heed full interrelating are critical to survival in a world that is now commonly perceived as uh full of surprises and so yeah this is something just by being on like the inside of a lot of these organizations including corporations a lot of times it's perceived as hey there's it's a Scary World out there we just need to get our information better to kind of understand what's actually happening so we don't have um any problems but I think the problem is that like the world is changing so much faster than it had been before where like this kind of Relentless approach of hey we're just going to get more we're going to gradually improve our processes to wash away this uncertainty that doesn't work as well anymore and I think this kind of former approach uh strategy is very much encapsulated by like Six Sigma design I don't know if you guys are familiar with that but it's kind of like a manufacturing idea of like hey we're just going to slowly and incrementally increase the Precision of our techniques and our processes because these any instruments you have Cascades across your whole supply chain and all of your

other processes so we just need to figure everything out and so everything will be perfect and we won't have any issues

um and that may be like somewhat of an illusion that you can tell yourself if things aren't changing that quickly and maybe that'll work for like some constrained amount of time but in the modern world it's hard really hard to have that illusion you have to have a different mental model of the world

um all right move on to the unless blue and Daniel had anything no all right um so the second piece of this is like yes the world is changing really quickly and that means that there's going to be more discontinuities and so he says our discounting of these are fundamentally a function of the cost of interacting with the outer World in form of generating and testing predictions and so Wix is not discontinuity it's really about like this gap between perceiving the world and saying okay

how are we going to take action how are we going to update our generative model accordingly accordingly and

what he's saying is that you know as the world becomes more complex change is much faster than ever did before there are more of these discontinuities but also the discontinuities are larger there's just more information to integrate and so like on one level you could say like this is really terrifying world's changing what do we do but he also sounds this optimistic note of

saying like well yes the modern world creates more discontinuities but it also creates more tools for navigating business continuities and so the first one he says is well you know is about distance remediation and decentralization so it's a lot easier to aggregate data and process that information and have backup plans in case plan A fails you have plans you know B through Z instead the other aspect of this is AI and so it's a lot easier to process bigger and deeper sets of data and so he's trying to sound this optimistic note that hey not All Is Lost there is a way to navigate through this uncertainty

I just have to say I love this meme as someone as someone who's like tried to like organize and restructure my life probably I don't know 500 000 times at this point

um like it's like do I just do the work or do I spend time like agonizing about how to like document and structure and like time plan my day into five minute blocks so like like I invest so much time in like some method right and then I'm just like okay well I methodized it and then I never used the method I just go back to doing the work so it's it's like this weird like all this time analyzing what like or dumping it into like moderating my efficiency is like wasting my efficiency

to connect that to the earlier discussion about the two senses of meta this time cost associated with strategy

and with consequences in the real world that time cost can be understood as a cognitive cost and something internal that's happening metacognitive how should I decide how to decide how to perceive this some of those are variables and knobs you can understand and change other ones are really subliminal and then also that time cost strategy Nexus of course exists for organizations and teams and all of these nested and complex interactions so we want to use active inference as an integrative principle because it helps us connect what's happening on either side of the Markov blanket which might be like a data set passed between two areas of a community or of an organization understanding how the emission model and the reception model work together

yeah to attach that a little more is that like

you know I was I've been a product Analyst at uh some like a couple of different startups and you actually see this in a real way where like sometimes you can get an okay answer by just taking like the like a super like simple average of a couple different categories or you could try to do something really clever and create like an app like you know an ml model or something over um to predict something and a lot of times it's like it doesn't the predictive power you gain doesn't match the time cost of that and by the time you created this really sophisticated

like publishable uh model the world has fundamentally changed so much that your model just like doesn't matter anymore uh and so like you're very like time-bounded and like how you can even like use the information you gather and the results you you glean so for example if you even if you have like a a checkout flow and you say oh yeah like uh with this button the checkout flow is 10 more effective than with this button well it could it will very likely be the change in six months the context of that checkout flow has changed so much that it's actually just kind of meaningless information and so like the time-boundedness of the information you collect and the perceptions you can form of that matter a lot

all right

um all right so the inference machine um so to survive the firm must become one cross hierarchically integrated inference machine that continuously builds predictions which are calling to be bottom of challenge

um and so I think this very much aligns with what we talked about before like agile methodology this kind of Lean Startup method of you build see if it worked you kind of like integrate those

um those perceptions and learn from it use that to ideate and something that is a creative act into the world not necessarily something that is fully informed by data

um and then you build and act on it

um and he uses this quote a few times both in the premise I think he actually opens the book with this uh he says there's a Carlo ravelli quote the search for knowledge is not nourished by certainty it is nourished by radical absence of certainty thinking born of curiosity revolt and change so this is actually very important because I've actually encountered this tension very often in a lot of organizations where like a lot of organization is one of the data driven and they don't actually another way to say this is this is the best product advice that was ever given any it was from CEO of my last companies and said there's two types of product work there's art and there's science and the science work is like you're like hey like we know this works we know we can speed up this checkout flow it might get like incrementally better we kind of can just keep going on there's like the exploit path but then there's art which is really just like not informed by the data and a lot of times organizations have trouble with this because it is truly a creative act and they're not comfortable taking that leap of faith and that's really what like you have to do to do anything uh meaningful and a lot of people are much more comfortable in the exploit path which is a lot easier to sell to your bosses it's a lot more comfortable and you have to really like CR like very consciously create a culture within your firm that like the

explore path is like even okay and something that can be rewarded um

all right anything that you guys want to add here before we on the next slide

all right um so something that I found of uh something I found that was interesting with

you know reading this preface and the first chapter was that like a lot of the themes are kind of emerging in a less technical Frameworks and also kind of like almost like pseudo-spiritual ideas about business management uh and so this is like a quote that was from the paper uh or from the book it says that firms are called to open up all four channels of listening listen from what you know what surprises you from the whole and what and what you and what you sense once you emerge the emerging home right and so if we kind of think about integrating information across different levels of the hierarchy like uh Jean was talking about like you can think about that from a more like rigorous academic perspective but there's like a lot of other books especially that come over the last five years that are conceiving the organization

um you know almost like a pseudo-spiritual lens where it's conceiving the organization not as like the extension of the CEO's ego but really as like this entity that needs to be nurtured and cultivated and understood on its own terms one of the most famous books about this is called

Reinventing organizations and this reading organizations introduce the idea of teal organizations these are like non-hierarchical heterogeneous organizations

um so in that book he says instead of trying to predict and control the future members of the organization are invited to listen in and understand what the organization wants to become what purpose it wants to serve which like on first reading like that can feel very like

pseudo-spiritual and kind of like hand wavy but is actually like a very real thing that can actually be modeled in a more rigorous way with activator and so

I think that's really exciting another way that this has been talked about has been uh in Brian J Robertson's book holography holocracy is another similar to like teal organizations it is a non-hierarchical form of governance and so he says each tension human being sense is a signpost telling us how the organization could evolve to better express its purpose

all right anything you guys want to add that

cool

cool so I don't know anything else that you guys want to add

um just thanks for being here and going over this uh stuff with me

um and is there anything else you're looking forward to or or what's next I'm really excited awesome oh just logistically how is the series going to

continue

sure so we'll finish one chapter a week

is the goal

um maybe like we're trying to still work

out the logistics of the holiday but I

think it looks consistent

um so one chapter of the book per week

something about that something like

along those lines and then a discussion

after each like section of the book so

there's part one part two part three

um and we will get in touch with the

author and see maybe if he is willing to

come and discuss some of these ideas

with us or and hopefully we'll have

other

um people that want to participate in

the discussions of the works

Tyler and then I'll give a thought

yeah I mean I think the thing I'm

looking forward to is you know so far

what I've been excited about is that

putting a a concrete framework behind

ideas that have so far been expressed in

very abstract kind of hand wavy terms

across you know different kind of

startup folklore and

um kind of business type businessy books

and so I'm curious as a Next Step about

how it can become a little more like

tangible and practical and like for

someone who is from the perspective like

being in an organization how do you how

would you actually design an

organization to embody pre-energy

governance

um and so I'm really excited to see how

that can become a little more concrete

and clear

cool the part I'm looking forward to is

continuing to explore that explore

exploit dialectic

both of those in the extreme

lead to failure in some way or at least

increase risk and so this frames our

strategizing as being like in some sort

of landscape with explore and exploit

which of course is really complex with

these nested organizations

and then you brought in the Art and

Science or um science and engineering or

just art and communication or whatever

other dialectic and then even within an

artist's day there's realities like

accounting and the mixing of the paint

and even the prompt engineering for

someone who's using those kinds of

artistic forms so it lets there be sort

of a light

and adaptive quote explore exploit

scenario happening within the context of

something that's a lot more meaningful

which is the integration of like art

science personal meaning Collective

meaning and so on without denying that

those kinds of explore exploit

attentional and strategic Dynamics must

happen also

cool well thanks for participating guys

and um we'll see you next time

bye y'all